

Skills Summary

Interpreting Correlation and Least-Squares Regression

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading a value out of the data

Rule: before any line is fitted, be sure which column is the explanatory variable x and which is the response y , and read values off the table or scatterplot in those units. Every interpretation later has to be phrased in those same units.

Example: A table records deliveries per day: Mon 4, Tue 7, Wed 9, Thu 12. How many on Wednesday?

Step 1. The question names a single category, so this is a direct read of that column, not a computation.

Step 2. Find the Wed column and read the count beneath it.

$\Rightarrow$ 9 deliveries

Try it: From the same table, what is the largest daily count?

check: 12

Skill 2 — Predicting and finding residuals

Rule: predicted value $\hat{y} = a + bx$; residual = $y - \hat{y}$ (actual minus predicted). A positive residual means the point sits *above* the line, a negative one below. Least squares chooses the line that makes the sum of the squared residuals as small as possible, which is why the residuals always sum to zero.

Example: For $\hat{y} = 12 + 2.5x$, predict y when $x = 6$.

Step 1. A prediction is a straight substitution into the fitted line — no data point is needed.

Step 2. $\hat{y} = 12 + 2.5(6) = 12 + 15$.

$\Rightarrow \hat{y} = 27$

Try it: The actual value at $x = 6$ turned out to be 30. Find the residual.

check: residual = 3

Skill 3 — What the slope and intercept mean

Slope: for each one-unit increase in x , the line predicts a change of b units in y . Say “is associated with”, not “causes”. **Intercept:** the predicted y when $x = 0$ — meaningful only if $x = 0$ lies inside the range of the data. Predicting outside that range is *extrapolation* and is not supported by the data.

Example: For $\hat{y} = 12 + 2.5x$, how much change does the line predict over 4 extra units of x ?

Step 1. The slope is a rate, so the predicted change is the slope times the change in x — the intercept plays no part.

Step 2. 2.5×4 .

$\Rightarrow$ predicted change = **10** units of y

Try it: Same line. What change does it predict over 10 extra units of x ?

check: 25

Skill 4 — Correlation and r^2

Formula:

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

r runs from -1 to 1 and measures the strength and direction of a *linear* pattern. r^2 is the fraction of the variation in y accounted for by that linear relationship with x . Neither says anything about cause: a lurking variable can drive both.

Example: A data set has $\sum(x - \bar{x})(y - \bar{y}) = 48$, $\sum(x - \bar{x})^2 = 20$, $\sum(y - \bar{y})^2 = 128$. Find r .

Step 1. All three sums the formula needs are given, so substitute rather than returning to the raw data.

Step 2. $r = \frac{48}{\sqrt{20 \cdot 128}} = \frac{48}{\sqrt{2560}}$.

$\Rightarrow r \approx \mathbf{0.9487}$, a strong positive linear association.

Try it: For the same three sums, find r^2 .

check: $0.9 = r^2$